

Porting the Umenberger–Göktoğan 6DOF paramotor model to the 0.33 kg PEEK robot

A change plan for `paramotor_mjcf/build_paramotor.py` → `paramotor.xml`

Reference: J. Umenberger and A. H. Göktoğan, “Guidance, Navigation and Control of a Small-Scale Paramotor,” ACRA 2012 ([pap151.pdf](#)).

1 The governing constraint: these are not scale models of each other

Nothing dimensional in Table 1 of the paper transfers to this robot, and no scaling law repairs that. The paper’s aircraft is 1.55 kg on a 2.15 m wing. This robot is 0.325 kg on a 1.0 m wing, with a 0.400 kg ceiling. Those two vehicles are not geometrically similar, not dynamically similar, and not Reynolds-similar — they differ in aspect ratio, in wing loading, in relative density and in mass distribution, independently and in different directions.

The paper’s value here is its **model structure** — the functional form of eqs. (8)–(22) — and its **identification method** (§3.2). Its numbers are worth only their order of magnitude, and several are worth less than that.

Every parameter in Table 1 falls into one of three classes, and the whole plan is organised around the distinction:

Class	Transfers? ra- me- ters
I. Dimensional	Never. The robot already knows all of these — from its own BOM, measured or budgeted. Read them out of the compiled model; do not scale the paper’s. A^B , A^P , A^F , b , c , d , $[s_{PB}]$, $[s_{FB}]$, $[s_{MF}]$, $[s_{CF}]$
II. Non-dimensional, geometry-driven	Provisionally , with an aspect-ratio correction. These are the real deliverable of the paper. C_D^P , C_L^P , $C_{D\alpha}^P$, C_{l_p} , C_{m_q} , C_{n_r} , χ
III. Non-dimensional, configuration-driven	No. Each depends on something this robot does differently: Reynolds number, line spacing, or brake cascade geometry. Structure transfers, values must be re-identified. C_D^F , $C_{D\alpha}^F$, C_{l_φ} , C_{m_0} , C_{m_α} , $C_{L\delta_a}$, $C_{D\delta_a}$, $C_{l\delta_a}$, $C_{n\delta_a}$

2 Similarity audit

All robot figures below are **measured from the compiled paramotor.xml**, not assumed. The composite inertia is accumulated over all 30 geoms about the system CG; the CG sits at $z = 0.1256$ m, i.e. 0.110 m above the pod CG and 0.415 m below the canopy CG.

	Paper	Robot	ratio
Class I – dimensional. Read the robot column; ignore the paper column.			
m [kg]	1.55	0.3254	0.210
b [m]	2.15	1.000	0.465
$\bar{c}$ [m]	0.54	0.1961	0.363
A^P [m ²]	1.16	0.1961	0.169
I_{xx} [kg·m ²]	0.336	0.02002	0.060
I_{yy}	0.292	0.01622	0.056
I_{zz}	0.109	0.005290	0.049
I_{xz}	−0.059	$−9.48 \times 10^{-5}$	0.002
ℓ_P (canopy CG above sys CG) [m]	1.066	0.4152	0.390
ℓ_F (pod CG below sys CG) [m]	0.149	0.1097	0.736
Class II – mass distribution, $i = I/(mb^2)$. These should match if the vehicles were similar.			
i_{xx}	0.04690	0.06153	1.312
i_{yy}	0.04075	0.04983	1.223
i_{zz}	0.01521	0.01626	1.069
i_{xz}	−0.008235	−0.000291	0.035
ℓ_P/b	0.4958	0.4152	0.837
ℓ_F/b	0.06930	0.1097	1.583
Aspect ratio b^2/A^P	3.985	5.100	1.280
Class III – the groups that decide whether identical coefficients give identical dynamics			
Wing loading m/A^P [kg/m ²]	1.336	1.660	1.242
Relative density $\mu = m/(\rho A^P b)$	0.5073	1.355	2.670
V_{trim} at $C_L = 0.6$ [m/s]	5.97	6.66	1.114
Froude $V^2/(gb)$	1.691	4.516	2.670
Reynolds ($\bar{c}$, V_{trim})	2.15×10^5	0.87×10^5	0.405
Time unit $t^* = b/2V$ [s]	0.180	0.0751	0.417

2.1 Reading the audit

The three bolded rows are the whole story.

I_{xz} is not a small number that got smaller – it is absent. The paper's $i_{xz} = -0.0082$ is 18% of i_{xx} : that is a pilot-carrying trike with a motor and cage slung well aft and below. The robot's $i_{xz} = -0.00029$ is 0.5% of i_{xx} – it is very nearly symmetric about its x - y plane, because it is a symmetric box with a pusher motor on the centreline. **Transcribing the paper's I_{xz} , at any scale factor, injects a roll-yaw inertial coupling this vehicle does not have.** This is the single most damaging number to copy.

$\mu = 2.67 \times$ and $\text{Fr} = 2.67 \times$ are the same fact. The robot is 2.67× denser relative to the air it flies in. Consequence, with an identical $C_{l_p} = -0.127$: the non-dimensional roll subsidence eigenvalue is

$$\hat{\lambda} = \frac{C_{l_p}}{8\mu i_{xx}}, \quad (1)$$

giving $\hat{\lambda} = -0.667$ for the paper and -0.190 for the robot — **3.5× weaker damping in non-dimensional time**. Dimensionally the gap narrows, because the small vehicle’s clock runs 2.4× faster:

Roll subsidence, same C_{l_p}	$\hat{\lambda}$	λ [1/s]	τ [s]
Paper	−0.667	−3.71	0.270
Robot	−0.190	−2.54	0.394

So the **dimensional** roll time constant is only 1.46× slower — but you reach that conclusion only by carrying μ and i_{xx} through. Anyone who transplants the coefficients and assumes the dynamics come with them will be wrong by 3.5× on every non-dimensional stability margin in the paper’s §4.

Re is 2.5× lower. 0.87×10^5 on a 125 μm single skin is a regime where C_{D_0} is dominated by laminar separation and is materially worse than at 2.15×10^5 . $C_{D_0}^P = 0.15$ is the least defensible transfer in the set.

Two ratios that are closer than expected, and are good news. Wing loading agrees to 24% and trim speed to 11%, so 6.0 m/s (V_{DESIGN}) remains the right linearisation airspeed. And ℓ_P/b agrees to 16% — the robot’s pendulum geometry is a reasonable facsimile.

A correction to MODEL_NOTES.md and to the previous revision of this plan. `LINE_HEIGHT_FRAC = 0.63` is the canopy **body origin** above the **pod origin**, and comparing it to the paper’s $1.066/2.15 = 0.496$ is an apples-to-oranges comparison — the paper’s figure is canopy CG above **system** CG. The arched panels hang below the canopy origin and the system CG sits 0.126 m above the pod origin, so the true robot figure is $0.4152/1.0 = 0.415$. The two rigs are 16% apart, not 27%, and in the opposite direction to what a naive reading suggests.

3 What actually has to change in the MJCF

The functional gap is not a matter of coefficients at all. MuJoCo’s ellipsoid fluid primitive (`fluidshape="ellipsoid"`, `fluidcoef="0.1 0.25 0.5 1.0 1.0"`) has no C_{L_0} , no C_{L_α} , and no stability derivatives. **It is a different function, not a different parameterisation of the same function.** No value of `CANOPY_FLUIDCOEF` reproduces eq. (12). The paper’s model has to be added as an explicit force callback, and MuJoCo’s own fluid model switched off.

3.1 Disable the built-in fluid model — or double-count everything

Mechanism	Where	Action
Ellipsoid fluid on panels	<code>class="skin"</code>	delete <code>fluidshape</code> , <code>fluidcoef</code>
Legacy inertia-box drag	implied by <code><option density viscosity></code> on every geom without <code>fluidshape</code>	set both to 0
Prop-disk opt-out hack	<code>fluidcoef="0 0 0 0 0"</code> on <code>prop_disk</code>	redundant once density is 0; remove

Carry $\rho = 1.225$ as a constant **in the callback**, not in `<option>`. Air density then becomes a parameter of the aero model, which is also what lets you sweep altitude later.

3.2 Do not touch the inertia

The previous revision of this plan proposed emitting an explicit `<inertial>` with the paper’s tensor geometrically scaled. **That was wrong and should not be done.** Against the compiled model it is off by:

	Geometric scaling $k_m k_L^2 = 0.0454$	Compiled model	error
I_{xx}	0.01526	0.02002	−24%
I_{yy}	0.01326	0.01622	−18%
I_{zz}	0.004950	0.005290	−6%
I_{xz}	+0.002680	−0.0000948	28× and wrong sign

The robot’s inertia is the accumulation of a traceable bill of materials. That is **better** information than any scaling of another aircraft’s tensor, and the only correct action is to leave MuJoCo to compute it. The paper’s tensor is used for exactly one thing: the replication check of §6.

3.3 Implement the aero layer in the paper’s frame, and convert once

The paper is forward–right–down; the MJCF is forward–left–up. The previous revision proposed a table of sign-flipped coefficients to type in. **Do not do this either** — it is where I got $C_{l_{\delta_a}}$ and $C_{n_{\delta_a}}$ backwards, and it is where anyone else will too.

Instead, convert the **state** in and the **force** out, and keep eqs. (8)–(22) verbatim in between:

$$T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad v^{\text{FRD}} = T v^{\text{FLU}}, \quad f^{\text{FLU}} = T f^{\text{FRD}}, \quad M^{\text{FLU}} = T M^{\text{FRD}} \quad (2)$$

$T = T^{-1} = T^T$, so the same matrix serves in both directions and there is one line of code to get right instead of fifteen table entries.

For cross-checking only, the correct signs are below. Note that δ_L and δ_R name **physical** brake lines, so $\delta_a = \delta_L - \delta_R$ is a frame-independent scalar and does **not** flip — which is what makes $C_{l_{\delta_a}}$ invariant and $C_{n_{\delta_a}}$ flip, not the other way round.

Quantity	FRD → FLU	Reason
L, p, φ	invariant	+roll is right-wing-down in both frames
$M, q, \theta, \alpha, \beta$	flip	y and z reverse
N, r, ψ	flip	
δ_a, δ_s	invariant	named by physical line, not by axis
$C_{l_p}, C_{l_\varphi}, C_{m_q}, C_{m_\alpha}, C_{n_r}$	invariant	$s_i s_j = +1$
C_{m_0}	flip	$s_M = -1$, unpaired
$C_{l_{\delta_a}}$	invariant	$s_L s_{\delta_a} = +1$
$C_{n_{\delta_a}}$	flip	$s_N s_{\delta_a} = -1$
I_{xz}	flip	$I' = T I T^T$

3.4 The remaining structural additions

Term	Equation, and what is missing today	Status
Parafoil L/D	(12)–(16) at P , offset $[s_{PB}]$ from the CG	replaces ellipsoid fluid
Fuselage drag	(8)–(11) at F . The pod has zero aero model today — <code>MODEL_NOTES.md</code> §10 records this.	new
Pure moments	(18): $C_{l_p}, C_{l_\varphi}, C_{m_q}, C_{m_0}, C_{m_\alpha}, C_{n_r}$. The only rotational damping today is <code>fluidcoef[2] = 0.5</code> , a blunt-body term, not a stability derivative.	new
Brake force + moment	(20)–(22), driven by a normalised δ (§4.6)	new
α clamp	not in the paper. (15) is linear with no stall; at $C_{L_\alpha} = 2$ the model reaches $C_L = 1.4$ by 30° and keeps going.	new, ours

Correction, verified against MuJoCo rather than assumed (`test_aero.py::test_xfrc_acts_at_com`): `xfrc_applied` acts at the body **centre of mass**, not at the body frame origin. Since the paper puts the parafoil aerodynamic centre at the parafoil mass centre and the fuselage drag at the fuselage mass centre, both forces land exactly where MuJoCo applies them. **No offset moment is required at all** in the two-body model, and the leverage moments of eq. (17) emerge through the tendons instead of being imposed. Only the pure moments of eq. (18) are applied explicitly.

3.5 Brakes: the input is not the same input

The MJCF actuates brakes geometrically — a 40 mm servo arm swinging through 3.0 rad, pulling a tendon anchored at the tip-panel trailing edge. The paper uses $\delta_{L/R} \in [0, 1]$. The bridge is a **calibration**, not a physical identity:

$$\delta = \text{clamp}\left(\frac{\ell(\theta_{\text{arm}}) - \ell_0}{\ell_{\text{max}} - \ell_0}, 0, 1\right), \quad \ell(\theta) = \sqrt{D^2 + r^2 - 2Dr \cos \theta} \quad (3)$$

with $r = \text{ARM_LEN} = 0.040$ m, D the pivot-to-anchor distance, ℓ_0 the rigged length including $\text{BRAKE_FREE} = 0.005$ m. The generator already computes Equation 3 for the tendon range; it only has to be exposed as a scalar.

The brake coefficients cannot be transferred under any scaling. Table 1 gives $C_{l_{\delta_a}} = +0.0021$; §3.2 identifies $C_{l_{\delta_a}} = -0.2959$ — a factor of 140 **and a sign flip**, on the same aircraft, in the same paper. The table’s nominal brake values are placeholders. The identified values are real, but they were identified for a brake cascade of length $d = 0.40$ m on a 2.15 m span ($d/b = 0.186$) spreading load across the trailing edge. This robot pulls **one tendon at one corner of one tip panel**. The authority distribution is not similar, so neither is the coefficient.

Transfer the **structure** of (20)–(22) and the **sign pattern**; identify $C_{l_{\delta_a}}$ and $C_{n_{\delta_a}}$ on this vehicle.

3.6 Actuator lags: use the robot’s own hardware, not the paper’s

The paper’s lags are its own hardware and are not vehicle-independent:

	Paper	This robot	Source
Servo transit	0.15 s/60°	0.09 s/60° @ 7.4 V	BD10BL-CAN datasheet, <code>itemized_spec.md</code> L123
$\omega_{\text{servo}} = 2.2/T_{\text{rise}}$	14.7 rad/s	24.4 rad/s	
τ_{servo}	0.0680 s	0.0409 s	
τ_{motor}	0.4545 s	identify	their prop+ESC; ours is far lighter

The robot’s servo is 1.7× faster. Its rotor inertia is $I_{\text{spin}} = 2.58 \times 10^{-5}$ kg·m², which will give a spin-up constant far below 0.45 s — take it from a bench step, do not copy 2.2 rad/s. Both go in as native first-order activation dynamics:

```
<position class="servo_pos" name="servo_pos_L" joint="arm_L"
    dyntype="filter" dynprm="0.0409"/>
```

3.7 Keep what the MJCF does better

The MJCF models propeller drag torque ($K_M/K_T = 0.0105$ m, via the `skydio_x2` gear trick) and carries a real `prop_spin` DOF so MuJoCo generates the gyroscopic moment natively ($H \approx 0.027$ N·m·s at full thrust). **The paper has neither term.** On a 0.33 kg vehicle that pitches and yaws under brake, these are the same order as the brake yaw moment. Keep them; document the divergence from the paper rather than removing it to match.

4 Implementation status

Sections 3.1, 3.3 and 3.4 are implemented and tested (32 assertions, `test_aero.py`). Sections 3.5 (brakes) and 3.6 (actuator lags) are deliberately untouched; `paramotor_aero.brake_wrench_frd()` is the hook and returns zero.

Section	Result	State
3.1	<code>density="0" viscosity="0"</code> ; no geom carries <code>fluidshape/fluidcoef</code> ; <code>CANOPY_FLUIDCOEF</code> deleted. Vehicle falls at exactly $g dt^2 n(n-1)/2$ with aero off — the discrete free-fall value, not $gt^2/2$.	done
3.3	Equations kept verbatim in FRD; $T = \text{diag}(1, -1, -1)$ applied once at the boundary. A guard refuses to run if MuJoCo’s fluid model is re-enabled.	done
3.4	Parafoil L/D , fuselage drag, six pure moments, α clamp with excursion reporting.	done
3.5	Brake hook present, returns zero.	not started
3.6	Actuator lags absent.	not started

4.1 What it does now

Quantity	Value	Check
Unpowered glide speed	5.47 m/s	audit predicts 6.7 at $C_L = 0.6$
Sink rate	1.69 m/s	
Glide ratio L/D	3.07	itemized_spec.md assumes 2–3
Powered climb at 1.5 N [†]	+0.79 m/s	
Fuselage drag at 6 m/s	0.340 N	was zero before; 11% of weight
Lift at $\alpha = 0$, 6 m/s	1.729 N	$= \frac{1}{2}\rho A^P V^2 C_{L_0}$ exactly
Energy drift at $\rho = 0$, 10 s	0.0	callback does no spurious work

[†] with the propeller drag reaction removed — see below.

4.2 Resolved: strip theory over the panels

The lumped single-force model could not produce a roll **restoring** moment, and that — not insufficient damping — was the cause of the powered spiral. Damping against a steady torque fixes the roll **rate**, so bank grows without bound; only a restoring term bounds it. Strip theory applies eqs. (12)/(15) to each of the fourteen panels separately, and because the canopy is arched each panel sees a different local incidence, so the roll physics falls out of the geometry:

	lumped	strip	note
C_{l_p}	−0.127	−0.221	1.74× the damping
C_{l_β}	0	−0.292 /rad	produced by nothing before
$\partial M/\partial \beta$	0	−1.26 N·m/rad	0.71° sideslip balances prop torque
total lift	1.729 N	1.727 N	preserved via arch recovery

At 0.5 N the lumped model rolls to 63° and dives; strip holds **0.1°** and glides. At 1.0 N it **climbs** with the propeller drag reaction left fully intact — the lumped model needed that torque deleted to manage the same thing. The hardware fact stays in the model, as §3.7 requires.

Arch recovery. An arched canopy lifts on its projected area, and strip theory reproduces that exactly from geometry — measured 0.7979 against the generator’s PROJ_FRACTION = 0.797. But the paper’s C_{L_0} is referenced to **flat** area on a wing that was already arched, so the loss is inside the coefficient. Applying the arch again costs 20% of the lift and a 12%-high trim speed. The factor (1.2533, computed from geometry) removes it from the coefficient so the geometry can supply it.

A correction to §4 of this plan. C_{l_φ} was set to zero on the argument that the suspension tendons produce the roll restoring moment structurally. They do not — they produce the **gravity** pendulum, canopy displaced relative to pod. The arc’s **aerodynamic** dihedral effect is a separate mechanism and was modelled by nothing at all. That was the direct cause of the spiral, and it is why the lumped mode still spirals under power.

4.3 Validated envelope, and one accepted limitation

The validated thrust range is 0 to about 1.0 N (T/W 0.31), and the spiral is gone across all of it. Above that the vehicle pitches up, the tension-only suspension goes fully slack (20 of 20 lines) and the canopy tumbles, with α reaching $\pm 180^\circ$ — far outside the $-8^\circ/+18^\circ$ envelope where a linear no-stall C_L means anything.

Established **not** to be a thrust-line offset (moving the propeller from $z = 0.010$ to the CG at $z = 0.126$ does not help) and **not** a step-input artifact (ramping with the §3.6 motor lag $\tau = 0.45$ s does not help). It needs a stall model and a rigging re-trim, both existing plan items. The lumped model was already failing above 0.5 N as a dive, so strip theory did not introduce this — it widened the usable range.

Accepted. test_aero.py::test_validated_thrust_envelope pins both edges — 1.0 N good, 1.8 N departs — so the boundary cannot drift unnoticed.

5 Coefficient disposition

Symbol	Table 1	Identified	Disposition for this robot
Class II — transfer, with an AR correction from 3.99 → 5.10			
$C_{L_0}^P$	0.4	—	Use 0.4. Single-skin reflex camber; weakly AR-dependent.
$C_{L_\alpha}^P$	2.0	—	2.08. Lifting line: invert at paper AR for $a_0 = 2.43$ ($e = 0.9$), re-evaluate at 5.10. A 4% change — defer if busy.
$C_{D_\alpha}^P$	1.0	—	0.92. Induced part $C_{L_\alpha}^2 / (\pi e \text{ AR})$ falls $0.355 \rightarrow 0.277$.
C_{l_p}	−0.1	−0.127	Use −0.127. Scales with C_{L_α} , which barely moves. But see Equation 1: same coefficient, 3.5× weaker non-dimensional damping.
C_{m_q}	−2.0	—	Use −2.0. Tail-less; set by ℓ_P , and ℓ_P/b agrees to 16%.
C_{n_r}	not tabulated	−0.0035	Use −0.0035. Note the paper omits it from Table 1 — a gap in the source.
χ	20°	—	Use 20° as the initial rigging angle, then trim it out. In the two-body MJCF this is emergent from line lengths, not a parameter.
Class III — do not transfer the value			
$C_{D_0}^P$	0.15	—	Identify. Re is 2.5× lower; also absorbs line drag, and this robot's line count and diameter differ.
$C_{D_0}^F, C_{D_\alpha}^F, A^F$	0.15, 1.0, 0.5 m ²	—	Replace. Only the product $C_{D_0}^F A^F$ is physical: the paper's is 0.075 m ² — a cage-and-pilot bluff body. The robot's pod is $0.140 \times 0.110 = 0.0154$ m ² frontal; use $A^F = 0.0154$, $C_{D_0}^F = 1.0$, giving 0.0154 m ² . Geometric scaling of A^F (0.108 m ²) is indefensible and was wrong in the previous revision.
C_{l_φ}	−0.05	−0.0055	Zero in the two-body model. This is the arc-plus-pendulum restoring moment; the suspension tendons already produce it structurally. Including it double-counts. Needed only in the rigid configuration of §6.
C_{m_0}, C_{m_α}	0.018, −0.2	—	Re-trim. Both are set by where the lines attach relative to the aerodynamic centre. The robot's A/B/C rows are at 0.060, −0.060, −0.090 in canopy x — its own rigging, not the paper's. Use as a starting guess, then trim to $C_m = 0$ at α_{trim} .
$C_{L_{\delta_a}}, C_{D_{\delta_a}}$	0.0001, 0.0001	—	Identify. Both are 10^{-4} placeholders in Table 1 and are not credible as given.
$C_{l_{\delta_a}}$	0.0021	−0.2959	Identify. 140× disagreement within the paper; different brake cascade. Keep the sign as a sanity target.
$C_{n_{\delta_a}}$	0.004	−0.0506	Identify. Same. Flip sign on entry to FLU if not using Equation 2.
d	0.40 m	—	Set from this robot's own brake span. Only $C_{l_{\delta_a}} b/d$ is physical.

6 Validation: replicate first, then port

The two configurations must be kept distinct, and the replication is not optional — it is the only way to know the aero layer is right before its outputs become unfalsifiable.

Configuration A — replication. A throwaway MJCF of the **paper's** aircraft: one rigid body, explicit `<inertial>` with $m = 1.55$ and `fullinertia="0.336 0.292 0.109 0 0.059 0"` (note I_{xz} flipped per Equation 2), Table 1 verbatim, $C_{l_\varphi} = -0.0055$ present. Its only purpose is to reproduce published figures.

#	Test	Reference
A1	Static, $\alpha^P = 0$, $V_P = 6$: assert $L = \frac{1}{2}\rho A^P V_P^2 C_{L_0} = 10.2$ N through P .	eq. (12)
A2	Trim at throttle 0.54 of 10 N: level flight, $V \approx 6$ m/s.	§5, Fig. 7

#	Test	Reference
A3	Throttle steps +0.1... + 0.3, -0.1, -0.2: assert $\dot{z} = 4.85, 4.83, 4.78, 4.75, 4.72$ m/s, $K_{\text{alt}} \approx 4.79$.	Fig. 7
A4	Step +0.2, pitch response: 36% overshoot, $\zeta = 0.31$, $\omega_n = 1.53$ rad/s.	Fig. 8, eq. (41)
A5	Numerically linearise about $\theta = 0$; assert the state matrix matches eq. (25) elementwise.	eq. (25)
A6	<code>ss2tf</code> ; assert $T_{\text{para}} = (6.177s^2 + 16.88s + 47.11)/(s^4 + 10.38s^3 + 30.29s^2 + 59.09s)$.	eq. (31)

A6 is the strongest available check: it exercises the inertia tensor, all six moment derivatives, the geometry vectors and Equation 2 simultaneously, and a sign error anywhere puts a pole in the wrong half-plane.

Configuration B – the robot. Same callback, robot parameters, MuJoCo-computed inertia, two bodies, $C_{l_\varphi} = 0$.

#	Test	Expected
B1	Energy: $\rho = 0$, no thrust, 60 s – total energy conserved to 10^{-6} .	proves the callback does no spurious work
B2	Free trim: assert $V \approx 6.7$ m/s at $C_L \approx 0.6$.	§2 audit
B3	Roll impulse: assert subsidence $\tau \approx 0.39$ s.	Equation 1
B4	Pendulum: release from 10° bank, no aero. Assert $T \approx 1.51$ s ($\omega = 4.17$ rad/s).	$\sqrt{mg\ell_P/(I_{xx} + m\ell_P^2)}$
B5	Sign test: left brake only $\rightarrow$ yaw left first , roll following. Causal order matters – skid steering, not roll steering.	§2.4
B6	C_{l_φ} cross-check: run B rigid with $C_{l_\varphi} = -0.0055$ and two-body with 0; assert the roll mode frequencies agree. Divergence means the tendon rigging is not producing the restoring moment.	§4.x

B4 is worth stating explicitly because it needs **no aerodynamics at all** and depends only on mass, inertia and geometry – it will catch an inertia or CG error before any coefficient is in play.

7 Order of work

1. **Strip.** `density="0" viscosity="0"`, remove `fluidshape / fluidcoef`. Confirm the vehicle now falls ballistically – proof nothing else was quietly making lift.
2. **Write the callback** as a pure function of $(\text{state}, \text{params}) \rightarrow (f, M)$, in FRD, with Equation 2 at the boundary. No MuJoCo imports beyond `mjData`.
3. **Configuration A**, Table 1 verbatim. Pass A1–A6. Do not proceed until A6 passes.
4. **Configuration B**. Swap the parameter dict, restore the two-body model, MuJoCo inertia, $C_{l_\varphi} = 0$. Pass B1–B6.
5. **Identify** $C_{D_0}^P, C_{m_0}, C_{m_\alpha}, C_{l_{\delta_a}}, C_{n_{\delta_a}}$ on the real vehicle. The paper’s §3.2 recursive weighted least-squares / Kalman method is directly reusable and is the most portable thing in the paper.

8 Code layout

```

build_paramotor.py      - CANOPY_FLUIDCOEF; density/viscosity -> 0
                        (inertia: NO CHANGE - leave MuJoCo to accumulate it)
paramotor_aero.py       NEW. Pure (state, params) -> (f, M) in FRD. Eqs. (8)-(22).
paramotor_params.py     NEW. PAPER_ACRA2012 and PEEK_1M dicts, identical keys.
check_paramotor.py      + tests A1-A6, B1-B6.

T = np.diag([1.0, -1.0, -1.0])      # FLU <-> FRD, self-inverse

def aero(model, data):
    data.xfrc_applied[:] = 0.0
    R = data.xmat[BODY].reshape(3, 3)
    v_b = T @ (R.T @ data.cvel[BODY, 3:])      # -> FRD body frame
    w_b = T @ (R.T @ data.cvel[BODY, :3])
    f, M = paramotor_aero.forces(v_b, w_b, phi, delta_L, delta_R, P)  # FRD
    data.xfrc_applied[BODY, :3] = R @ (T @ f)    # -> FLU -> world
    data.xfrc_applied[BODY, 3:] = R @ (T @ M)

mujoco.set_mjcb_passive(aero)

```

9 Minimum change set

1. `<option density="0" viscosity="0">`; delete `fluidshape/fluidcoef` from `class="skin"` and `prop_disk`. Delete `CANOPY_FLUIDCOEF` — it has no image under this mapping.
2. Add `dyntype="filter"` with $\tau_{\text{servo}} = 0.0409$ s from **this robot's** datasheet, not the paper's 0.0680 s. Identify τ_{motor} .
3. Add the `mjcb_passive` callback implementing eqs. (8)–(22) in FRD.
4. Introduce sixteen new coefficients — **seven** taken from the paper ($C_{L_0}^P, C_{L_\alpha}^P, C_{D_\alpha}^P, C_{l_p}, C_{m_q}, C_{n_r}, \chi$), **nine** to be identified on this vehicle.
5. Take $A^P, b, c, [s_{PB}], [s_{FB}], [s_{MF}]$ and the full inertia from the compiled model. **Change none of them to match the paper.**